

ORDER MANAGEMENT SYSTEM

WEEK 4

1.Database Design

1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order Date	Date	NOT NULL
Total Amount	NUMBER(10,2)	NOT NULL

1.2 Order Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Quantity	INT	NOT NULL
Unit Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2. Table Relationship

Customer → Orders

- Customer.Customer_ID → Primary Key
- Orders.Customer_ID → Foreign Key
- Relationship: One-to-Many (1:M)
- One customer can place multiple orders.

Orders → Order_Details

- Orders.Order_ID → Primary Key
- Order_Details.Order_ID → Foreign Key
- Relationship: One-to-Many (1:M)
- One order can have multiple order details.

Product → Order_Details

- Product.Product_ID → Primary Key
- Order_Details.Product_ID → Foreign Key

- Relationship: One-to-Many (1:M)
- One product can have multiple order details.

3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     CONSTRAINT fk_orders_customer  
7         FOREIGN KEY (Customer_ID)  
8         REFERENCES Customer(Customer_ID)  
9 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Order_Details (  
2     Order_Detail_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER,  
4     Product_ID NUMBER,  
5     Quantity NUMBER NOT NULL,  
6     Unit_Price NUMBER(10,2) NOT NULL,  
7     Subtotal NUMBER(10,2) NOT NULL,  
8     CONSTRAINT fk_orderdetails_order  
9         FOREIGN KEY (Order_ID)  
10        REFERENCES Orders(Order_ID),  
11     CONSTRAINT fk_orderdetails_product  
12        FOREIGN KEY (Product_ID)  
13        REFERENCES Product(Product_ID)  
14 );
```

Table created.

4.Order Data Insertion

```
SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES (205, 105, TO_DATE('14-SEP-2026', 'DD-MON-YYYY'), 2750.00);
```

1 row created.

```
SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES (206, 106, TO_DATE('15-SEP-2026', 'DD-MON-YYYY'), 1950.00);
```

1 row created.

```
SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES (207, 107, TO_DATE('16-SEP-2026', 'DD-MON-YYYY'), 3400.00);
```

1 row created.

```
SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES (208, 108, TO_DATE('17-SEP-2026', 'DD-MON-YYYY'), 1250.00);
```

1 row created.

5.Order_Deatils Data Insertion

```
SQL> INSERT INTO Order_Details
  2  (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3  VALUES (6, 205, 102, 1, 350.00, 350.00);
```

1 row created.

```
SQL> INSERT INTO Order_Details
  2  (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3  VALUES (7, 206, 103, 2, 450.00, 900.00);
```

1 row created.

```
SQL> INSERT INTO Order_Details
  2  (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3  VALUES (8, 207, 104, 3, 299.00, 897.00);
```

1 row created.

```
SQL> INSERT INTO Order_Details
  2  (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3  VALUES (9, 208, 105, 2, 399.00, 798.00);
```

1 row created.

6.Order Modification Operations

6.1 Modify Order Amount

```
SQL> UPDATE Orders
  2  SET Total_Amount = 2900.00
  3  WHERE Order_ID = 205;
```

1 row updated.

```
SQL> SELECT *
  2  FROM Orders
  3  WHERE Order_ID = 205;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT
205	105	14-SEP-26	2900

6.2 Modify Order Date

```
SQL> UPDATE Orders
  2  SET Order_Date = TO_DATE('18-SEP-2026', 'DD-MON-YYYY')
  3  WHERE Order_ID = 206;
```

1 row updated.

7. Order_Details Modification

```
SQL> UPDATE Order_Details
  2  SET Quantity = 3,
  3      Subtotal = 1350.00
  4  WHERE Order_Detail_ID = 7;
```

1 row updated.

8. Customer Order History Report

```

SQL> SELECT
  2     c.Customer_ID,
  3     c.First_Name,
  4     c.Last_Name,
  5     o.Order_ID,
  6     o.Order_Date,
  7     od.Product_ID,
  8     od.Quantity,
  9     od.Unit_Price,
10     od.Subtotal
11 FROM Customer c
12 JOIN Orders o
13     ON c.Customer_ID = o.Customer_ID
14 JOIN Order_Details od
15     ON o.Order_ID = od.Order_ID
16 ORDER BY c.Customer_ID, o.Order_Date;

```

CUSTOMER_ID FIRST_NAME

-----					ORDER_ID	ORDER_DAT
LAST_NAME						

PRODUCT_ID	QUANTITY	UNIT_PRICE	SUBTOTAL			

105 Rahul						
K	102	1	350	350	205	14-SEP-26
106 Sneha						
P	103	3	450	1350	206	18-SEP-26

CUSTOMER_ID FIRST_NAME

-----					ORDER_ID	ORDER_DAT
LAST_NAME						

PRODUCT_ID	QUANTITY	UNIT_PRICE	SUBTOTAL			

107 Arun						
T	104	3	299	897	207	16-SEP-26

9. Customer-Wise Order Summary

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.First_Name,
  4     c.Last_Name,
  5     COUNT(o.Order_ID) AS Total_Orders,
  6     SUM(o.Total_Amount) AS Total_Amount
  7 FROM Customer c
  8 JOIN Orders o
  9     ON c.Customer_ID = o.Customer_ID
 10 GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 11 ORDER BY c.Customer_ID;
```

CUSTOMER_ID FIRST_NAME

LAST_NAME	TOTAL_ORDERS	TOTAL_AMOUNT
101 Harini	1	2500
105 Rahul	1	2900
106 Sneha	1	1950

CUSTOMER_ID FIRST_NAME

LAST_NAME	TOTAL_ORDERS	TOTAL_AMOUNT
107 Arun	1	3400
108 Divya	1	1250